11

The Jordan Canonical Form

Historical Note

Chapter 10 left this theorem owed, and also left its attribution unsettled. Weierstrass published the theory of elementary divisors in 1868, which determines everything below; Jordan's *Traité des substitutions* of 1870 gave a canonical form, but for linear substitutions over a finite field, in a book whose subject was permutation groups and the solvability of equations. The complex statement that everyone now calls the Jordan form is closer to Weierstrass's result than to Jordan's, and historians of the period have said so at length. The name stuck anyway. Frobenius added a third account in 1878, the rational canonical form, which has the advantage of needing no eigenvalues and therefore working over any field at all. One warning belongs in this paragraph rather than at the end: beautiful as it is, the Jordan form cannot be computed in floating point, for a reason Section 11.7 makes precise. It is a tool for proving things.

Theorem 10.7 said exactly when a matrix is diagonalisable: when every geometric multiplicity equals its algebraic multiplicity. When some geometric multiplicity falls short there are not enough eigenvectors to make a basis, and Chapter 10 stopped there. This chapter asks what to do instead, and the answer is that one can always get within a single superdiagonal of diagonal.

11.1 The obstruction, and the plan

Recall the shear $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ of Example 9.17: the eigenvalue 1 has algebraic multiplicity 2 and geometric multiplicity 1, so there is one eigendirection where two are needed. The missing vector is not missing entirely. Although no second v satisfies $(A - \mathbf{1})v = \mathbf{0}$, the vector e_2 satisfies

$$(A - \mathbf{1})e_2 = e_1, \qquad (A - \mathbf{1})^2 e_2 = \mathbf{0},$$

so it is killed by the *second* power. That is the whole idea: when eigenvectors run out, admit the vectors killed by higher powers, and organise them into chains.

Definition 11.1 (Generalised eigenvectors)**.** *Let T be an operator on V and λ a scalar. A* generalised eigenvector *for λ is a nonzero v with $(T-\lambda\,\mathrm{id})^j v = \mathbf{0}$ for some $j \geq 1$. The* generalised eigenspace *is*

$$G_\lambda = \ker(T-\lambda\,\mathrm{id})^n, \qquad n = \dim V.$$

Lemma 11.2 (The kernels stabilise)**.** *For any operator S on an n-dimensional space,*

$$\ker S \subseteq \ker S^2 \subseteq \ker S^3 \subseteq \cdots,$$

and once two consecutive terms agree the chain is constant thereafter. In particular it is constant from step n onwards, so $\ker S^n = \ker S^{n+1} = \cdots$.

Proof. If $S^j v = \mathbf{0}$ then $S^{j+1}v = \mathbf{0}$, which gives the inclusions. Suppose $\ker S^k = \ker S^{k+1}$ and let $v \in \ker S^{k+2}$. Then $S^{k+1}(Sv) = \mathbf{0}$, so $Sv \in \ker S^{k+1} = \ker S^k$, so $S^k(Sv) = \mathbf{0}$ and $v \in \ker S^{k+1}$. Hence $\ker S^{k+2} = \ker S^{k+1}$, and induction continues it. Until the chain stops it grows by at least one dimension each step, and dimensions are bounded by n, so it has stopped by step n. $\square$

11.2 Triangularisation, and Cayley–Hamilton

We need one structural fact that costs nothing beyond Chapter 9.

Lemma 11.3 (Triangularisation over $\mathbb{C}$)**.** *Every $A \in M_n(\mathbb{C})$ is similar to an upper triangular matrix.*

Proof. Induct on n, the case $n = 1$ being trivial. By Theorem 9.9 there is an eigenvalue λ with eigenvector v; extend v to a basis of $\mathbb{C}^n$ by Theorem 6.14(e) and let P_1 have those vectors as columns, v first. Then $P_1^{-1}AP_1$ has first column λe_1, so it is $\left(\begin{smallmatrix} \lambda & b^t \\ 0 & A_1 \end{smallmatrix}\right)$. Apply the inductive hypothesis to A_1, writing $P_2^{-1}A_1P_2 = T_1$ upper triangular, and conjugate by $\left(\begin{smallmatrix} 1 & 0 \\ 0 & P_2 \end{smallmatrix}\right)$; the result is $\left(\begin{smallmatrix} \lambda & b^tP_2 \\ 0 & T_1 \end{smallmatrix}\right)$, which is upper triangular. $\square$

This is Schur's theorem with the orthogonality removed, and it is available now because it needs no inner product: just an eigenvector and a basis extension.

Theorem 11.4 (Cayley–Hamilton)**.** *Every square matrix satisfies its own characteristic equation: $p_A(A) = 0$.*

Proof. The statement is unchanged by similarity, since $p_{P^{-1}AP} = p_A$ by Proposition 9.6 and $q(P^{-1}AP) = P^{-1}q(A)P$ for any polynomial q. So by Lemma 11.3 we may assume A is upper triangular with diagonal $\lambda_1, \dots, \lambda_n$, and then $p_A(x) = (-1)^n \prod_i (x - \lambda_i)$ by Proposition 9.18(c).

Let $W_j = \operatorname{span}\{e_1, \dots, e_j\}$, with $W_0 = \{\mathbf{0}\}$. Triangularity says $Ae_j \in W_j$, and the jth diagonal entry is λ_j, so $(A - \lambda_j \mathbf{1})e_j \in W_{j-1}$; since $(A - \lambda_j \mathbf{1})W_{j-1} \subseteq W_{j-1}$ as well, we get

$$(A - \lambda_j \mathbf{1})\, W_j \;\subseteq\; W_{j-1}.$$

Applying the factors from right to left,

$$(A - \lambda_1 \mathbf{1}) \cdots (A - \lambda_n \mathbf{1})\, V \subseteq (A - \lambda_1 \mathbf{1}) \cdots (A - \lambda_{n-1} \mathbf{1})\, W_{n-1} \subseteq \cdots \subseteq W_0 = \{\mathbf{0}\}. \qquad \square$$

Theorem 11.5 (Primary decomposition)**.** *Let T act on a complex space V of dimension n with distinct eigenvalues $\lambda_1, \dots, \lambda_k$ of algebraic multiplicities $m_1, \dots, m_k$. Then*

$$V = G_{\lambda_1} \oplus \cdots \oplus G_{\lambda_k},$$

each G_{λ_i} is invariant under T, and $\dim G_{\lambda_i} = m_i$. Moreover $G_{\lambda_i} = \ker(T - \lambda_i \operatorname{id})^{m_i}$.

Proof. Write $p(x) = (-1)^n \prod_i (x - \lambda_i)^{m_i}$ and $q_i(x) = \prod_{j \neq i} (x - \lambda_j)^{m_j}$. The polynomials $q_1, \dots, q_k$ have no common root, so they are collectively coprime and there are polynomials r_i with $\sum_i r_i q_i = 1$. Substituting T,

$$\sum_i r_i(T) q_i(T) = \operatorname{id}, \qquad \text{so} \qquad v = \sum_i v_i, \quad v_i := r_i(T) q_i(T) v.$$

Each v_i lies in $K_i := \ker(T - \lambda_i)^{m_i}$, because $(T - \lambda_i)^{m_i} q_i(T) = \pm p(T) = 0$ by Theorem 11.4 and $r_i(T)$ commutes with everything in sight. So the spaces $K_1, \dots, K_k$ span V.

Invariance is clear, since T commutes with $(T - \lambda_i)^{m_i}$. Note also that if $v \in K_j$ then $q_i(T)v = \mathbf{0}$ for every $i \neq j$, since q_i carries the factor $(x - \lambda_j)^{m_j}$; so $v = \sum_i r_i(T) q_i(T) v = r_j(T) q_j(T) v$, that is, $r_j(T) q_j(T)$ acts as the identity on K_j and as zero on every other K_i. Applying it to a vanishing sum $\sum_i v_i = \mathbf{0}$ with $v_i \in K_i$ therefore returns $v_j = \mathbf{0}$, for each j in turn. Hence $V = K_1 \oplus \cdots \oplus K_k$.

On K_i the operator T satisfies $(T - \lambda_i)^{m_i} = 0$, so λ_i is its only eigenvalue there and its characteristic polynomial on K_i is $\pm(x - \lambda_i)^{\dim K_i}$. Because the decomposition is direct and each summand is invariant, multiplying these over i recovers p, which forces $\dim K_i = m_i$.

It remains to identify K_i with $G_{\lambda_i} = \ker(T - \lambda_i)^n$. Certainly $K_i \subseteq G_{\lambda_i}$. Conversely take $v \in G_{\lambda_i}$ and write $v = \sum_j v_j$ with $v_j \in K_j$. For $j \neq i$ the operator $T - \lambda_i$ is invertible on K_j, since its only eigenvalue there is $\lambda_j - \lambda_i \neq 0$, so $(T - \lambda_i)^n v_j = \mathbf{0}$ forces $v_j = \mathbf{0}$. Hence $v = v_i \in K_i$, and the decomposition is the stated one. $\square$

11.3 Nilpotent operators

By Theorem 11.5 the whole problem reduces to one generalised eigenspace, on which $N = T - \lambda\,\mathrm{id}$ satisfies $N^m = 0$. So everything now depends on understanding nilpotent operators, and for those there is a complete and rather pretty answer.

Definition 11.6 (Jordan chain). *A chain of length k for a nilpotent N is a list $u,\ Nu,\ N^2u,\ \dots,\ N^{k-1}u$ with $N^k u = \mathbf{0}$ and $N^{k-1}u \neq \mathbf{0}$. Its last member is an eigenvector, since N sends it to zero.*

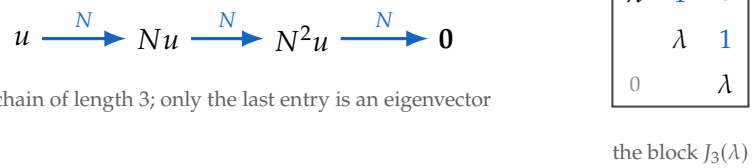


Figure 11.1. A chain and the block it produces. Reading the chain from the right, each basis vector is sent by T to λ times itself plus the previous one, which is exactly a λ on the diagonal and a 1 above it. A chain of length 1 is an ordinary eigenvector and gives a 1×1 block, so diagonalisable matrices are the case where every chain has length one.

Theorem 11.7 (Nilpotent operators are sums of chains). *Let N be nilpotent on a finite-dimensional V. Then V has a basis consisting of the members of finitely many chains.*

Proof. Induct on $\dim V$. If $N = 0$ every nonzero vector is a chain of length one and any basis will do.

Otherwise let $W = \operatorname{Im} N$, which satisfies $\dim W < \dim V$ because N is not injective. Now $N(W) \subseteq W$ and N restricted to W is nilpotent, so by the inductive hypothesis W has a basis made of chains

$$w_i,\ Nw_i,\ \dots,\ N^{k_i-1}w_i, \qquad i = 1, \dots, s,$$

with $\sum_i k_i = \dim W$. Each w_i lies in $\operatorname{Im} N$, so choose u_i with $Nu_i = w_i$; then

$$u_i,\ Nu_i,\ \dots,\ N^{k_i}u_i$$

is a chain of length $k_i + 1$ in V. Its last member $N^{k_i}u_i = N^{k_i-1}w_i$ lies in $\ker N$, and these s vectors are independent, being part of a basis of W. Extend them to a basis of $\ker N$ by adjoining $z_1, \dots, z_t$, each of which is a chain of length one, where $t = \dim \ker N - s$.

These vectors are independent. Suppose $\sum_i \sum_{j=0}^{k_i} c_{ij} N^j u_i + \sum_l d_l z_l = \mathbf{0}$ and apply N. Every z_l dies, as does every term with $j = k_i$, leaving

$$\sum_i \sum_{j=0}^{k_i - 1} c_{ij}\, N^j w_i = \mathbf{0},$$

a relation among a basis of W; so $c_{ij} = 0$ whenever $j < k_i$. What remains is $\sum_i c_{i,k_i} N^{k_i} u_i + \sum_l d_l z_l = \mathbf{0}$, a relation among a basis of $\ker N$, so the rest of the coefficients vanish too.

They span. Their number is $\sum_i (k_i + 1) + t = \dim W + s + (\dim \ker N - s)$, which is $\dim W + \dim \ker N = \dim V$ by Corollary 7.12. An independent set of that size is a basis, by Theorem 6.14(c). □

11.4 The theorem

Definition 11.8 (Jordan block). *$J_k(\lambda)$ is the $k \times k$ matrix with λ on the diagonal, 1 on the superdiagonal and 0 elsewhere.*

Theorem 11.9 (Jordan canonical form). *Every $A \in M_n(\mathbb{C})$ is similar to a block diagonal matrix*

$$J = J_{k_1}(\mu_1) \oplus J_{k_2}(\mu_2) \oplus \cdots \oplus J_{k_r}(\mu_r),$$

where each μ_i is an eigenvalue of A and $\sum_i k_i = n$. The list of blocks is unique up to their order.

Proof. Existence: decompose $\mathbb{C}^n = \bigoplus_\lambda G_\lambda$ by Theorem 11.5. On G_λ the operator $N = A - \lambda \mathbf{1}$ is nilpotent, so Theorem 11.7 supplies a basis of chains; in that basis N has ones on the superdiagonal within each chain and zeros elsewhere, so $A = N + \lambda \mathbf{1}$ has the stated block form, as Figure 11.1 shows. Collecting the bases over all λ gives a basis of $\mathbb{C}^n$.

Uniqueness follows from Theorem 11.10 below, which recovers the number of blocks of each size from the ranks of powers of $A - \lambda \mathbf{1}$, and those ranks are unchanged by similarity. □

Figure 11.2 shows what the resulting matrix looks like: diagonal apart from a few ones directly above the diagonal, with distinct blocks free to carry the same eigenvalue.

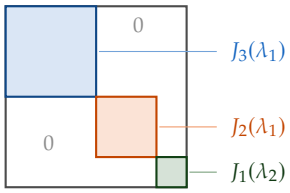


Figure 11.2. A Jordan form with three blocks. Everything outside the blocks is zero, so the matrix is diagonal except for a few ones directly above the diagonal. Distinct blocks may carry the same eigenvalue; when they do, the number of them is that eigenvalue's geometric multiplicity and the total of their sizes is its algebraic multiplicity, which is Theorem 9.16 made explicit.

11.5 Reading the blocks off the ranks

Theorem 11.10 (Block counts from ranks)**.** *Fix an eigenvalue λ and write $r_j = \operatorname{rank}(A - \lambda \mathbf{1})^j$, so $r_0 = n$. Then the number of Jordan blocks for λ of size at least j is*

$$r_{j-1} - r_j,$$

and the number of size exactly j is $r_{j-1} - 2r_j + r_{j+1}$. In particular the total number of blocks for λ is $n - r_1$, the geometric multiplicity.

Proof. Rank is unchanged by similarity, so compute in the Jordan basis, where $(A - \lambda \mathbf{1})^j$ acts block by block. On a block $J_k(\mu)$ with $\mu \neq \lambda$ the matrix $J_k(\mu) - \lambda \mathbf{1}$ is triangular with nonzero diagonal, hence invertible, so that block contributes k to every r_j and drops out of the differences. On a block $J_k(\lambda)$ the matrix is the nilpotent shift, whose jth power has rank $\max(k - j, 0)$. Hence

$$r_{j-1} - r_j = \sum_{\text{blocks for } \lambda} \big[\max(k - j + 1, 0) - \max(k - j, 0)\big],$$

and each bracket is 1 when $k \geq j$ and 0 otherwise, which is the first claim. Subtracting the count for $j + 1$ from the count for j gives the second. □

The bookkeeping is easier drawn than stated. Figure 11.3 puts one column per block, its height the size of that block, so that the number of columns is the geometric multiplicity and the number of dots the algebraic multiplicity.

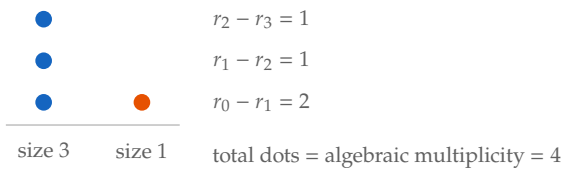


Figure 11.3. The dot diagram of Theorem 11.10, drawn for Example 11.11. One column per Jordan block, its height the block's size; the number of columns is the geometric multiplicity and the number of dots the algebraic multiplicity. Counting instead by rows, the jth row from the bottom holds one dot for every block of size at least j, and that count is $r_{j-1} - r_j$. The two ways of counting the same dots are the two halves of the theorem.

Example 11.11 (Recovering the blocks)**.** *Let A be 4×4 with the single eigenvalue 3, and suppose the ranks of $(A - 3\,\mathbf{1})^j$ are found to be*

$$r_0 = 4, \qquad r_1 = 2, \qquad r_2 = 1, \qquad r_3 = 0.$$

The number of blocks is $r_0 - r_1 = 2$. The number of size at least 2 is $r_1 - r_2 = 1$, and of size at least 3 is $r_2 - r_3 = 1$, while $r_3 - r_4 = 0$ stops the count. So one block has size at least 3 and no block has size 4: the blocks are $J_3(3)$ and $J_1(3)$, and

$$J = J_3(3) \oplus J_1(3).$$

Checking with the second formula: blocks of size exactly 1 number $r_0 - 2r_1 + r_2 = 4 - 4 + 1 = 1$; of size exactly 2, $r_1 - 2r_2 + r_3 = 0$; of size exactly 3, $r_2 - 2r_3 + r_4 = 1$. The sizes $3 + 1$ total 4, as they must.

Caution

Multiplicities alone do not determine the form. A 4×4 matrix with a single eigenvalue of geometric multiplicity 2 could be $J_3 \oplus J_1$ or $J_2 \oplus J_2$: both have two blocks and both total 4. The ranks separate them, since $r_2 = 1$ in the first case and $r_2 = 0$ in the second. This is why Theorem 11.10 needs every power and not just the first.

11.6 What it is for

Definition 11.12 (Minimal polynomial)**.** *The* minimal polynomial m_A *is the monic polynomial of least degree with* $m_A(A) = 0$.

Proposition 11.13. $m_A(x) = \prod_i (x - \lambda_i)^{d_i}$, *where* d_i *is the size of the* largest *Jordan block for* λ_i. *Consequently* A *is diagonalisable if and only if* m_A *has no repeated root.*

Proof. A polynomial annihilates A exactly when it annihilates every Jordan block, and $(J_k(\lambda) - \lambda \mathbf{1})^j = 0$ precisely when $j \geq k$, since that matrix is the nilpotent shift. So the least exponent that works for λ_i is the largest block size, and factors for distinct eigenvalues cannot interfere. Every d_i equals 1 exactly when every block is 1×1, which by Theorem 11.9 and Proposition 10.2 is diagonalisability. □

Example 11.14 (Powers of a block)**.** *Write* $J_k(\lambda) = \lambda \mathbf{1} + N$ *with* N *the nilpotent shift, so* $N^k = 0$. *Since* $\lambda \mathbf{1}$ *and* N *commute, the binomial theorem applies and terminates:*

$$J_k(\lambda)^m = \sum_{j=0}^{k-1} \binom{m}{j} \lambda^{m-j} N^j.$$

The entries therefore behave like $m^{k-1}\,|\lambda|^m$ *for large* m*: a polynomial in* m *times a geometric factor. Since a polynomial cannot outrun a geometric decay,* $J_k(\lambda)^m \to 0$ *exactly when* $|\lambda| < 1$*, whatever the block size. This is the fact Chapter 18 relies on when it says that the Jordan form contributes polynomial factors in the step count which do not affect whether an iteration converges.*

Key Idea

The Jordan form settles the classification question that Chapter 7 raised and Chapter 10 could only half answer. Two matrices are similar, meaning they are the same operator in different bases, if and only if they have the same Jordan blocks. So the list of blocks is a complete invariant: it is everything about an operator that does not depend on the choice of basis, and the trace, determinant, rank, characteristic polynomial and minimal polynomial are all functions of it.

11.7 Why it is never computed

Caution

The Jordan form is not computable in floating point, and the reason is not that the algorithms are poor. Consider

$$A_\varepsilon = \begin{pmatrix} 1 & 1 \\ 0 & 1+\varepsilon \end{pmatrix},$$

the matrix from the Caution in Section 10.4. For every $\varepsilon \neq 0$ it has distinct eigenvalues, so its Jordan form is diagonal with two blocks of size one. At $\varepsilon = 0$ it is a single block of size two. The Jordan structure therefore changes discontinuously at a point that no finite-precision computation can distinguish from its neighbours: rounding A_0 perturbs it by something of order $\varepsilon_{\text{mach}}$, and that is enough to change the answer completely.

So a program cannot decide block sizes from computed data, and Theorem 11.10 does not help, because deciding the rank of a computed matrix requires deciding which small singular values are really zero. What is computed instead is the Schur form of Theorem 14.5, which is unitary, stable, and available for every matrix. The Jordan form is for proofs.

Figure 11.4 measures how bad the discontinuity is rather than merely asserting it. Filling the empty corner of a single 2×2 block with ε moves the eigenvalues to $\pm\sqrt{\varepsilon}$, a line of slope $\frac{1}{2}$ on a log-log plot against the slope of one a well conditioned problem would give: at $\varepsilon = 10^{-8}$ the eigenvalues move by 10^{-4}, ten thousand times further.

Insight

It is worth asking what the Jordan form is really telling us, because the answer is not what the statement suggests.

Read naively, the theorem says every operator is nearly diagonal. Read honestly, it says the opposite: it exhibits exactly how far an operator can be from diagonal, and the obstruction is a single integer per eigenvalue, the size of the largest block. Everything a diagonalisable operator does is captured by its eigenvalues; a Jordan block adds one further behaviour, a nilpotent shift that no eigenvalue can see. That is why e^{tJ} acquires polynomial factors in t, why the transient of Chapter 73 exists at all, and why Chapter 31 found the square-root sensitivity it did.

The deeper lesson is about what a classification is for. The Jordan form classifies operators up to similarity completely, and it is almost never computed, because

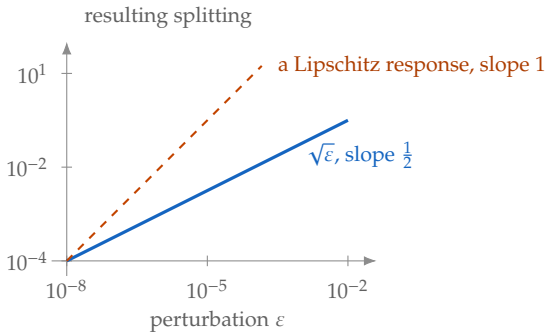


Figure 11.4. What a Jordan block does to a perturbation. The matrix $\left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)$ has 0 as a double eigenvalue and only one eigenvector; filling in its empty corner with ε gives eigenvalues $\pm\sqrt{\varepsilon}$. On a log-log plot that is a line of slope one half, and the gap against the dashed line of slope one is the whole point: at $\varepsilon = 10^{-8}$ a Lipschitz response would move the eigenvalues by 10^{-8} and the true answer moves them by 10^{-4}, ten thousand times further. No condition number explains it, because a defective matrix has no eigenvector basis for one to be computed from. For an $n \times n$ block the exponent is $1/n$, and Chapter 31 makes the general statement.

the block structure is decided by ranks of powers and rank is discontinuous. So here is a complete answer to a natural question that is also useless as an algorithm, and the two facts are not in tension: the theorem is a statement about what *exists*, and the numerical difficulty is a statement about what can be *observed*. Much of numerical linear algebra is the business of replacing exact classifications that cannot be computed with approximate ones that can, and Theorem 14.5, the unitary triangularisation, is the replacement for this one.

Open Problem 11.15 (The Deligne and Simpson problem)**.** *Fix conjugacy classes $C_1, \dots, C_k$ in $GL_n(\mathbb{C})$, that is, fix k Jordan forms.* When do there exist matrices $A_i \in C_i$ with $A_1 A_2 \cdots A_k = \mathbf{1}$ and no common invariant subspace?

What is known. *A necessary condition is immediate: taking determinants gives $\prod_i \det C_i = 1$. It is nowhere near sufficient. Crawley-Boevey solved the problem completely in the case where the classes are semisimple, and more generally whenever a certain root system condition holds, by translating the question into one about representations of a quiver and applying Kac's theorem [167]. The general case, with arbitrary Jordan structure, is open.*

Why the Jordan form is the whole difficulty. *If every C_i is a class of diagonalisable matrices, the constraint is about eigenvalues and dimensions of eigenspaces, and dimension*

counting gets a long way. As soon as a class contains a nontrivial Jordan block, the class is no longer closed: its closure contains smaller classes, and the product of closures is not the closure of the product. The problem is hard exactly to the extent that Chapter 11's classification is finer than a list of eigenvalues.

Where the question came from. *Differential equations. A linear system on the punctured sphere with regular singularities at k points has a monodromy representation, and the local behaviour at each puncture is a conjugacy class; asking which local behaviours are globally realisable is exactly the problem above. It is the modern form of Hilbert's twenty-first problem.*

Where a reader might start. *Take $k = 3$, $n = 2$, and all three classes equal to that of $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Can three unipotent 2×2 matrices with a single Jordan block each multiply to the identity without a common eigenvector? Work it out; the answer is instructive either way, and the same computation in 3×3 is already beyond hand calculation.*

Exercises

Exercise 11.16. *Find the Jordan form of each matrix, and a basis of chains realising it.*

(a) $\left(\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right)$

(b) $\left(\begin{smallmatrix} 5 & 0 \\ 0 & 5 \end{smallmatrix}\right)$

(c) $\left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$

Exercise 11.17. *A 6×6 matrix has the single eigenvalue 2, and the ranks of $(A - 2\mathbf{1})^j$ for $j = 1, 2, 3, 4$ are $3, 1, 0, 0$. Determine the Jordan form, draw its dot diagram, and state the minimal polynomial.*

Exercise 11.18. *Give two 4×4 matrices with the same characteristic polynomial and the same minimal polynomial that are nevertheless not similar.* Hint: five blocks are possible for a single eigenvalue of multiplicity four; find two of them that agree on the largest block size.

Exercise 11.19. *Verify Cayley–Hamilton directly for $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$ by computing p_A and then $p_A(A)$. Then use $p_A(A) = 0$ to express A^{-1} as a polynomial in A.*

Exercise 11.20. *How many similarity classes of nilpotent 5×5 complex matrices are there?* Hint: a nilpotent Jordan form is determined by the sizes of its blocks, which are a partition of 5.

Exercise 11.21. *Let N be nilpotent. Show that $\mathbf{1} + N$ is invertible and write its inverse as a finite sum. Deduce that every eigenvalue of $\mathbf{1} + N$ equals 1.*

Exercise 11.22. *Using Example 11.14, compute $e^{J_2(\lambda)t}$ explicitly, and use it to solve $x' = J_2(0)x$ with $x(0) = (0, 1)$. Why does a solution grow like t even though the eigenvalue is zero?*

Exercise 11.23. *For $A = \left(\begin{smallmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{smallmatrix}\right)$, compute $\ker(A - 3\mathbf{1})$, $\ker(A - 3\mathbf{1})^2$ and G_3, and confirm Lemma 11.2 and Theorem 11.5.*

Exercise 11.24. *A real matrix may have no real Jordan form, because its eigenvalues may be complex. What is the Jordan form over $\mathbb{C}$ of the rotation $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and why does no real similarity produce it?* The real substitute, which pairs conjugate eigenvalues into 2×2 rotation blocks, is called the real Jordan form.

References and further reading

The historical question of what Jordan actually proved, and how his name became attached to the complex canonical form, has been examined carefully in the recent literature on nineteenth-century algebra, and the short answer is that Weierstrass has the better claim. Axler [37] reaches the Jordan form through invariant subspaces and nilpotent operators without determinants or the characteristic polynomial, which is cleaner than our route and worth reading alongside it; our proof of Theorem 11.7 is the standard one by induction on the image. Treil [610] gives the rational canonical form as well, which is the version that survives over fields where the characteristic polynomial does not split. Banica [51], our baseline, treats the Jordan form as preparation for the operator-theoretic material of its later parts. On the numerical side the definitive remark is Golub and Van Loan's [263]: the Jordan form is a valuable theoretical tool that no one should attempt to compute, and Trefethen and Bau [607] explain the discontinuity in terms of the pseudospectrum, which is the modern way to describe what goes wrong.